

RESEARCH

Open Access



Influencing factors and changing trends of depressive symptoms among middle and junior high school students in Eastern China from 2019 to 2023: a cross-sectional study

Xingchen Zhu^{1,2}, Na An^{1*}, Zaixiang Tang^{2*}, Jianping Huang¹, Qingxin Ren¹ and Yue Wu¹

Abstract

Background The emergence of the COVID-19 pandemic led to increased levels of depression among adolescents in 2019. However, the situation regarding adolescent depression in the later stages of the pandemic remains unclear. This study investigated depressive symptoms status and changing trends among middle and junior high school students from 2019 to 2023.

Methods Data were collected from 9,920 middle and junior high school students between 2019 and 2023 via multistage stratified cluster sampling. Students were administered an online survey, completed the Chinese version of the Center for Epidemiologic Studies Depression Scale, and answered questions regarding basic information, such as birth date. Univariate and multivariate analyses were performed to examine the factors that influenced depressive symptoms. Furthermore, Trend χ^2 Test and Logistic regression was employed to analyze the changing trends of depressive symptoms among the participants.

Results Of the students, 14.5% and 5.8% could have and had definitely experienced depressive symptoms, respectively. Junior high school students (AOR = 1.409, 95%CI = 1.164–1.706) had a higher risk of definite depressive symptoms. Males (possible depressive symptoms: AOR = 0.803, 95%CI = 0.716–0.902; definite depressive symptoms: AOR = 0.829, 95%CI = 0.695–0.988) were less likely to experience depressive symptoms. Families with parental absence (single-parent families, reconstituted families, and families where both parents are absent), less than one hour of outdoor activity and less than eight hours of sleep, increased the risk of both possible and definite depressive symptoms. Detection rate of depressive symptoms among middle school students followed an inverted U-shape trend; it rose in 2020 and 2021 and declined in 2022 and 2023. Detection rates of depressive symptoms among junior high school students in 2020, 2022, and 2023 decreased compared with 2019 ($p < 0.05$).

*Correspondence:

Na An
kuailedeanna@163.com
Zaixiang Tang
tangzx@suda.edu.cn

Full list of author information is available at the end of the article



© The Author(s) 2024. **Open Access** This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

Conclusion Educational stage, sex, family type, and duration of outdoor activities and sleep were factors that influenced depressive symptoms among middle and junior high school students. A proportion of middle school students with depressive symptoms exhibited an increasing and decreasing trend from 2019 to 2023. Students exposed to risk factors require additional attention, with timely early intervention to mitigate the impact of environmental stress and prevent the further development of depressive symptoms into a more severe and less manageable state.

Keywords Mental health, Adolescents, Depressive symptoms, COVID-19, China

Background

Approximately 1 in 7 adolescents aged 10–19 years worldwide suffer from psychiatric disorders, accounting for 13% of the total disease burden in this age group [1]. With the rapid development of economic and social sectors, the prevalence of mental health problems and psychiatric disorders among children and adolescents is increasing [2, 3]. It has become an important public health and social concern related to the future growth of countries and nations [4]. Mental health problems in adolescents, such as depression and anxiety, affect academic performance, reduce quality of life, increase family burden, and lead to higher rates of substance use [5]. A survey report on the mental health status of adolescents in 2022, the “Report on the Development of Mental Health in China (2021–2022),” indicated that approximately 14.8% were at risk of depression. Furthermore, 10.8% and 4.0% experienced mild and severe depressive risk, respectively [6]. Therefore, depression among students remains a focal point that warrants attention from society, schools, and parents, and should command serious concern.

Emergence of the COVID-19 pandemic, in late December 2019, affected economic and social development and the general population’s physical and mental health [7]. The pandemic may have long-term effects on mental health [8–10]. However, most current studies have focused on the period between 2019 and 2022, during which participants were still impacted by COVID-19 and its control measures. During the COVID-19 outbreak, based on expert advice, Chinese students mainly reduced their risk of infection through home isolation and online learning, which led to changes in outdoor activity time, sleep time, and permanent household population [11]. The COVID-19 pandemic posed various psychological problems for different populations, including students [12, 13]. By the end of 2022, China’s COVID-19 policy changed, and the virus was classified as a Class B infectious disease, which allowed economic and social practices to gradually return to normal. Lockdown measures were relaxed for middle and junior high school students, and they returned to campus. A longitudinal study in Shandong Province investigated 1,170 middle school students at three time-points: May 2021, November 2021, and May 2022. The study concluded that during the later

stages of the pandemic, depressive and anxious emotions among Chinese adolescents did not worsen; rather, they showed a trend of improvement [14]. To the best of our knowledge, no long-term study has evaluated the changing trends in depressive symptoms among adolescents before and after the COVID-19 pandemic, especially from 2022 onwards. The impact of the COVID-19 pandemic is far greater and more widespread than that of previous epidemics, such as severe acute respiratory syndrome and Middle East respiratory syndrome. Therefore, long-term monitoring of its effects on adolescent mental health as well as the formulation and implementation of intervention plans are essential [15]. Understanding the changes in the detection rates of depressive symptoms among Chinese adolescents is crucial for the early development of effective prevention and intervention strategies.

This study aimed to investigate the influencing factors and changing trends of depressive symptoms among middle and junior high school students, in the Chinese city of Nantong over the past five years, to explore the short- and long-term effects of the COVID-19 pandemic on adolescent mental health. Furthermore, we aimed to provide suggestions for mental health interventions among middle and junior high school students. Nantong, a medium-sized city located in the Eastern coastal region, boasts a favorable geographical position. According to data from the 2020 population census, Nantong had a population of 7.72 million people, and its regional gross domestic product reached 1.18133 trillion yuan in 2023. Its educational system is highly competitive, with significant emphasis placed on secondary education and college entrance examination (Gaokao) performance, making it one of the top cities in China for secondary education. Therefore, investigating the mental health of middle and junior high school students in Nantong has reference value and considerable significance.

Methods

Study design

To monitor the mental health of students and the influencing factors on their health, we conducted a cross-sectional study from 2019 to 2023 in Nantong. The survey was conducted between September and October each year, except in November of 2022. Multi-stage stratified

cluster sampling was employed to select an urban district and suburban county in Nantong as the primary sampling points. Based on the results of the 2017 Ministry of Education School Survey, two middle and high schools were selected from the urban district, two middle schools and one high school were selected from the suburban county. Within each school, the sampling was stratified by grade level, with at least two classes selected from each grade to ensure that a single selection would not be less than 80 students. Both schools and classes were selected via simple random sampling. All students in the selected classes were included as participants when the following conditions were met: they were present at school on the day of the survey, aged between 12 and 18 years, local residents, had no history of psychiatric disorders except for depressive symptoms, had no severe diseases such as tumors that could affect health, and had no communication barriers. A total of 10,171 individuals were included. After 251 duplicates and data for 89 individuals without depressive symptoms were excluded, the final sample size included 9,831 students. A G Power analysis indicated that 531 participants would be adequate for a moderate effect (0.3, $\alpha=0.05$, $1-\beta=0.80$). Our study was conducted under strict confidentiality, and the data were to be used solely for scientific research. The survey was conducted in a designated computer room where all students answered the questions online. During the investigator training, we conducted a pilot test. Trained investigators provided instructions on how to complete the questionnaire and emphasized confidentiality. We obtained informed consent from all students and their parents or guardians and this study was approved by the Ethics Committee of the

Nantong City Disease Prevention and Control Center (approval no. 2023004). Figure 1 illustrates the sample selection process.

Survey of the baseline characteristics

Survey data included demographic characteristics, duration of outdoor activities and sleep, and the Center for Epidemiologic Studies Depression Scale (CES-D). The CES-D, designed by Radloff [16] to assess the degree of current depressive symptoms, focused on depressive affect or mood. It has been used for comparison of cross-sectional survey results at different time points. The CES-D is an acceptable, reliable scale and sensitive to changes. As we aimed to identify as many depressive symptoms as possible, we accepted the drawback of the high false positive rate of the CES-D. The CES-D is widely used to assess depressive symptoms among adolescents [17]; its reliability and structural validity in Chinese adolescent samples have been confirmed [18–20]. The CES-D comprised 20 items and five factors: depressive mood, positive mood, somatic symptoms, activity retardation, and interpersonal relationships. Responses were rated on a four-point Likert scale, where 1 (none or occasionally), 2 (sometimes), 3 (often or half the time), and 4 (most of the time or continuously) were rated as 0, 1, 2, and 3 points. Questions 4, 8, 12, and 16 were reverse-scored. The total score ranged from 0 to 60, and 0–15, 16–27, and <28 points indicated no, possible, and definite depressive symptoms, respectively.

Furthermore, we included additional questions to assess lifestyle and household types. Duration of daily outdoor activities was defined as follows: during the past

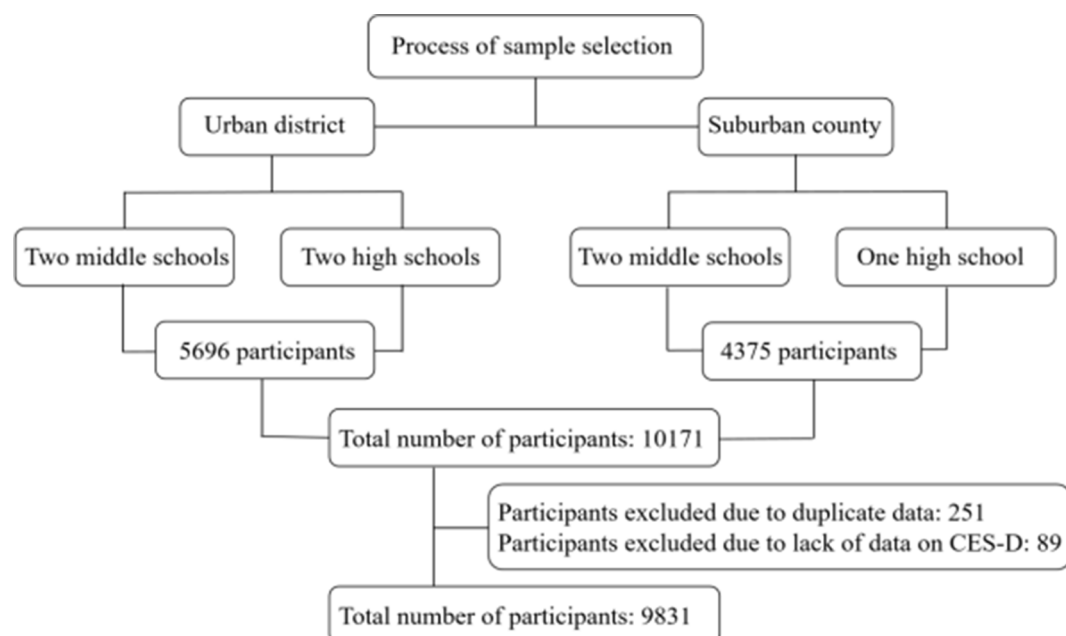


Fig. 1 Process of sample selection

Table 1 General characteristics of middle and junior high school students ($N=9831$)

Item	Number	Percentage
Educational stage		
Middle school	5501	56.0
Junior high school	4330	44.0
Sex		
Male	5181	52.7
Female	4650	47.3
Students' mental status		
Normal	7840	79.7
Possible depressive symptoms	1424	14.5
Definite depressive symptoms	567	5.8
Year		
2019	2072	21.1
2020	2035	20.7
2021	1732	17.6
2022	1916	19.5
2023	2076	21.1
Total	9831	100.0

week, time spent in outdoor activities during the day was less than 1 h, 1–2 h (excluding 2), 2–3 h (excluding 3), or 3 h or more. Sleep duration was divided into <5 h, 5–6 h (excluding 5), 6–7 h (excluding 7), 7–8 h (excluding 8), and ≥ 8 h. The permanent household population was divided into five categories according to students' self-reports: nuclear, large, single-parent, reconstituted, and family with both parents absent.

Data analysis

Statistical analysis was performed using SPSS version 26.0. Qualitative and quantitative data were expressed as cases (percentages) and mean $\pm$ standard deviation, respectively. Chi-squared tests were used to analyze the differences in the detection rate of depressive symptoms among different groups and investigate the relationship between depressive symptoms and demographic traits among middle school and junior high school students. Pairwise comparisons between multiple groups were corrected via the Bonferroni adjustment. Values with $p < 0.05$ in the univariate analysis were included in the multivariate logistic regression model as independent variables (forward). Multivariate logistic regression was employed to identify the factors that influenced possible and definite depressive symptoms among middle school students. Then, we reclassified depressive symptoms from a three-category classification to a binary classification: existence of depressive symptoms (including possible and definite depressive symptoms) and no symptoms, and conducted a logistic regression analysis. Stratified analyses were conducted according to sex (male, female) and educational stage (middle school, junior high school). Furthermore, trend X^2 and logistic regression analyses

Table 2 Depressive symptoms among middle and junior high school students from 2019–2023. ($N=9831$)

Year	Normal	Possible depressive symptoms	Definite depressive symptoms
2019	1629(78.6)	320(15.4)	123(5.9)
2020	1595(78.4)	329(16.2)	111(5.5)
2021	1337(77.2)	281(16.2)	114(6.6)
2022	1583(82.6)	233(12.2)	100(5.2)
2023	1696(81.7)	261(12.6)	119(5.7)
Total	7840(79.7)	1424(14.5)	567(5.8)

were conducted to examine changes in the detection rate of depressive symptoms during the entire study period, adjusted for these influencing factors. The significance level was set at $p < 0.05$.

Results

Descriptive analysis

In total, 9,920 middle school and junior high school students from Nantong were enrolled between 2019 and 2023. After 89 individuals with missing CES-D results were excluded, data from 9,831 (99.10%) students were analyzed. Cronbach's alpha coefficients of the scale for 2019–2023 were 0.860, 0.868, 0.871, 0.877, and 0.873, respectively, which indicated good reliability.

Sex ratio and proportion of junior high school to middle school students did not exhibit statistically significant differences from 2019 to 2023 (Supplementary Table 1). Of the participants, 14.5% and 5.8% had possible and definite depressive symptoms during the study period, respectively (Tables 1 and 2).

Univariate analysis

Educational stage, sex, family style, outdoor activities, and sleep duration were all factors that influenced depressive symptoms among middle and junior high school students. The proportion of middle school students who may have had depressive symptoms was lower than that of junior high school students; however, no significant difference was observed in the proportion of those who definitely had depressive symptoms. Females had a higher proportion of both possible and definite presence of depressive symptoms compared with males. Students from nuclear families and large families had a higher proportion of depressive symptoms compared with other family structures. Detection rate of depressive symptoms was higher among those with less than 1 h of outdoor activity duration compared with 1–3 h. Furthermore, students with sleep durations that exceeded 7 h had a lower detection rate of depressive symptoms (Table 3).

Table 3 Univariate Chi-square test of depressive symptoms among middle and junior high school students ($N=9831$)

Item	Normal	Possible depressive symptoms	Definite depressive symptoms	P
Educational stage				
Middle school	4494(81.7) ^a	692(12.6) ^a	315(5.7) ^a	< 0.001
Junior high school	3346(77.3) ^b	732(16.9) ^b	252(5.8) ^a	
Sex				
Male	4227(81.6) ^a	678(13.1) ^a	276(5.3) ^a	< 0.001
Female	3613(77.7) ^b	746(16.0) ^b	291(6.3) ^b	
Family style				
Nuclear family	4063(81.5) ^a	667(13.4) ^a	255(5.1) ^a	< 0.001
Large family	2114(81.2) ^a	359(13.8) ^{a, b}	129(5.0) ^a	
Single-parent family	1225(75.8) ^b	276(17.1) ^c	115(7.1) ^b	
Reconstituted family	177(69.4) ^b	52(20.4) ^c	26(10.2) ^b	
Family with both parents absent	261(70.0) ^b	70(18.8) ^{b, c}	42(11.3) ^b	
Outdoor activities (hours)				
< 1	2054(73.4) ^a	523(18.7) ^a	221(7.9) ^a	< 0.001
1–2	3515(82.0) ^b	567(13.2) ^b	203(4.7) ^{b, c}	
2–3	1216(84.4) ^b	167(11.6) ^b	57(4.0) ^c	
≥ 3	1055(80.7) ^b	167(12.8) ^b	86(6.6) ^{a, b}	
Sleep duration (hours)				
< 5	79(52.0) ^a	35(23.0) ^{a, b}	38(25.0) ^a	< 0.001
5–6	570(60.2) ^a	246(26.0) ^b	131(13.8) ^b	
6–7	2650(76.8) ^b	579(16.8) ^a	220(6.4) ^c	
7–8	2293(84.0) ^c	336(12.3) ^c	101(3.7) ^d	
≥ 8	2248(88.1) ^d	228(8.9) ^d	77(3.0) ^d	
Total	7840(79.7)	1424(14.5)	567(5.8)	9831(100)

Notes The number range includes the initial number. For example, 1–2 includes the number 1. ^{a–d} different letters in the same column indicate significant statistical difference ($p < 0.05$) as indicated by Bonferroni correction

Multivariate logistic regression

Educational stage, sex, family style, outdoor activities, and sleep duration were included in a multiple regression model. To ensure the robustness of our findings, a multicollinearity test was done with all Variance Inflation Factors (VIF) below 2 (range: 1.016–1.846): no multicollinearity was found among the variables. For possible depressive symptoms, coming from a single-parent family (AOR=1.359, 95%CI=1.161–1.589), reconstituted family (AOR=1.837, 95%CI=1.324–2.548), or family with both parents absent (AOR=1.710, 95%CI=1.289–2.268) and having less than 8 h of sleep (<5 h: AOR=3.903, 95%CI=2.549–5.976; 5–6 h: AOR=3.759, 95%CI=3.027–4.667; 6–7 h: AOR=1.961, 95%CI=1.643–2.341; 7–8 h: AOR=1.372, 95%CI=1.143–1.648) were identified as risk factors; being male (AOR=0.803, 95%CI=0.716–0.902)

and having more than 1 h of daily outdoor activities (1–2 h: AOR=0.747, 95%CI=0.652–0.855; 2–3 h: AOR=0.666, 95%CI=0.549–0.808; ≥3 h: AOR=0.785, 95%CI=0.646–0.955) were protective factors. For definite depressive symptoms, being a middle school student (AOR=1.409, 95%CI=1.164–1.706), coming from a single-parent family (AOR=1.434, 95%CI=1.134–1.814), reconstituted family (AOR=2.231, 95%CI=1.431–3.480), or family with both parents absent (AOR=2.443, 95%CI=1.698–3.516) and having less than 8 h of sleep (<5 h: AOR=13.710, 95%CI=8.659–21.710; 5–6 h: AOR=7.289, 95%CI=5.315–9.996; 6–7 h: AOR=2.678, 95%CI=2.020–3.551; 7–8 h: AOR=1.372, 95%CI=1.009–1.866) were identified as risk factors, being male (AOR=0.829, 95%CI=0.695–0.988) and having 1–3 h of outdoor activities (1–2 h: AOR=0.656, 95%CI=0.534–0.807; 2–3 h: AOR=0.557, 95%CI=0.409–0.759) were identified as protective factors (Table 4).

The results of the logistic regression analysis for binary-classified depressive symptoms were consistent: sex, family type, duration of outdoor activities, and sleep duration emerged as significant influencing factors (Table 5). Stratified analyses were performed based on sex and educational stage. For sex, educational stage, family type, duration of outdoor activities, and sleep duration remained significant factors. For educational stage, sex and having 6–8 h of sleep were no longer significant for junior high school students (Supplementary Tables 2 and 3).

Annual trend analysis

Comparison of the detection rates of depressive symptoms from 2019 to 2023 revealed that with the outbreak of COVID-19, the detection rates of depressive symptoms among middle school students increased in 2020 and early 2021 compared with 2019. Following changes in pandemic policies after 2022, detection rates among middle school students decreased in 2022 and 2023 (2019–2021: trend $X^2=4.487$, $p=0.034$; 2021–2023: trend $X^2=18.348$, $p<0.001$). From 2019 to 2023, depressive symptoms trends among middle school students exhibited an inverted U-shape. No such change was observed among junior high school students (Figs. 2 and 3).

After adjustment for sex, family type, duration of outdoor activities, and sleep duration, we used detection rate of depressive symptoms and year as the dependent variable and factor, respectively, to employ a logistic regression model and analyze the changes in the detection of depressive symptoms among middle and junior high school students. Among middle school students, compared with 2019, the detection rate of depressive symptoms increased in 2020 and 2021. Compared with 2021, it decreased in 2022 and 2023. Meanwhile, in 2023, the detection rate in this population was lower than that in

Table 4 Multivariate logistic regression analysis of the impact on possible and definite depressive symptoms ($N=9831$)

Students' mental status	Item	β	Standard error	Wald	P	AOR	AOR (95%CI)	
Possible depressive symptoms	Educational stage							
	Middle school vs. junior high school	-0.044	0.065	0.451	0.502	0.957	0.843	1.087
	Sex							
	Male vs. female	-0.219	0.059	13.718	<0.001	0.803	0.716	0.902
	Family style							
	Large family vs. ref*	0.045	0.072	0.387	0.534	1.046	0.908	1.204
	Single-parent family vs. ref	0.306	0.080	14.649	<0.001	1.359	1.161	1.589
	Reconstituted family vs. ref	0.608	0.167	13.256	<0.001	1.837	1.324	2.548
	Family with both parents absent vs. ref	0.536	0.144	13.863	<0.001	1.710	1.289	2.268
	Outdoor activities (hours)							
	1–2 vs. < 1	-0.292	0.069	17.949	<0.001	0.747	0.652	0.855
	2–3 vs. < 1	-0.407	0.099	17.003	<0.001	0.666	0.549	0.808
	≥ 3 vs. < 1	-0.241	0.100	5.835	0.016	0.785	0.646	0.955
	Sleep duration (hours)							
	< 5 vs. ≥ 8	1.362	0.217	39.249	<0.001	3.903	2.549	5.976
Definite depressive symptoms	5–6 vs. ≥ 8	1.324	0.110	143.751	<0.001	3.759	3.027	4.667
	6–7 vs. ≥ 8	0.674	0.090	55.612	<0.001	1.961	1.643	2.341
	7–8 vs. ≥ 8	0.316	0.093	11.478	0.001	1.372	1.143	1.648
	Educational stage							
	Middle school vs. junior high school	0.343	0.098	12.337	<0.001	1.409	1.164	1.706
	Sex							
	Male vs. female	-0.188	0.090	4.380	0.036	0.829	0.695	0.988
	Family style							
	Large family vs. ref	-0.047	0.114	0.169	0.681	0.954	0.764	1.192
	Single-parent family vs. ref	0.361	0.120	9.057	0.003	1.434	1.134	1.814
	Reconstituted family vs. ref	0.803	0.227	12.523	<0.001	2.231	1.431	3.480
	Family with both parents absent vs. ref	0.893	0.186	23.132	<0.001	2.443	1.698	3.516
	Outdoor activities (hours)							
	1–2 vs. < 1	-0.421	0.105	16.005	<0.001	0.656	0.534	0.807
	2–3 vs. < 1	-0.584	0.158	13.752	<0.001	0.557	0.409	0.759
	≥ 3 vs. < 1	-0.034	0.139	0.061	0.804	0.966	0.735	1.269
	Sleep duration (hours)							
	< 5 vs. ≥ 8	2.618	0.234	124.664	<0.001	13.710	8.659	21.710
	5–6 vs. ≥ 8	1.986	0.161	152.008	<0.001	7.289	5.315	9.996
	6–7 vs. ≥ 8	0.985	0.144	46.914	<0.001	2.678	2.020	3.551
	7–8 vs. ≥ 8	0.316	0.157	4.071	0.044	1.372	1.009	1.866

Notes *Nuclear family was considered as reference (ref). AOR: Adjusted odds ratio

2019. Among junior high school students, the detection rates in 2020, 2022, and 2023 decreased compared with 2019. Among middle and junior high school students, the detection rates in 2022 and 2023 decreased compared with 2019 and 2021 (Tables 6 and 7; Figs. 2 and 3).

Discussion

Adolescent depressive symptoms are a significant health issue. In this study, the detection rate of depressive symptoms among middle and junior high school students in Nantong was 20.3%. Depression among middle and junior high school students remains a focal point that deserves attention from society, schools, and parents. This study explored the factors that influenced depressive symptoms among middle school students and inferred a

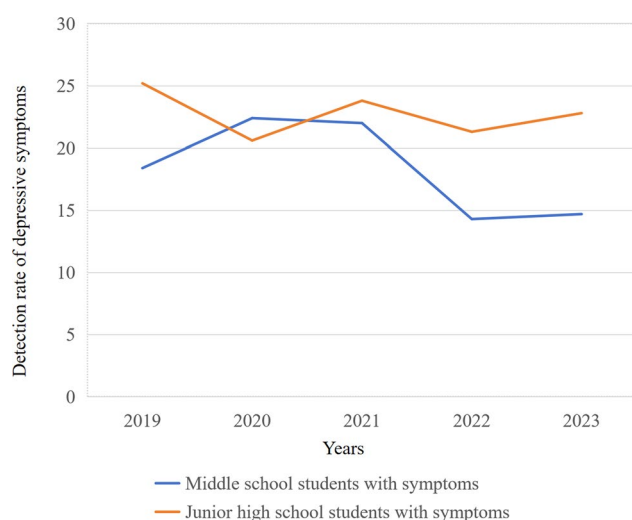
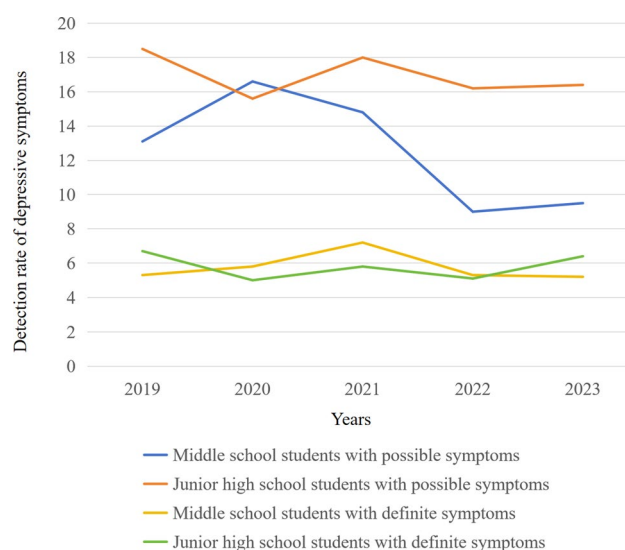
trend in the detection rate of depressive symptoms over the past five years. Thereby, we provide additional clues for mental health interventions targeting middle and junior high school students.

We considered a CES-D score of 16 and 28 as indicative of possible and definite depressive symptoms, which accounted for 14.5% and 5.8% of the sample, respectively. This was essentially consistent with the report from the developers, who stated that a score of 16 and 28 corresponded to the 80th and 95th percentile, respectively [16]. Thus, our determination of depressive symptoms boundaries was appropriate. We found that the proportion of students with depressive symptoms was similar to that reported by Duan et al. [21] (22.3%) and Xie et al. [11] (22.6%); however, it was lower than that reported

Table 5 Multivariate logistic regression analysis of the impact on depressive symptoms among middle school and junior high school students ($N=9831$)

Item	β	Standard error	Wald	P	AOR	AOR (95%CI)	
Educational stage							
Middle school vs. junior high school	0.064	0.057	1.259	0.262	1.066	0.953	1.193
Sex							
Male vs. female	-0.210	0.052	16.142	<0.001	0.811	0.732	0.898
Family style							
Large family vs. ref*	0.021	0.064	0.105	0.746	1.021	0.901	1.157
Single-parent family vs. ref	0.322	0.071	20.743	<0.001	1.380	1.201	1.585
Reconstituted family vs. ref	0.665	0.146	20.716	0.000	1.945	1.461	2.591
Family with both parents absent vs. ref	0.651	0.123	27.843	0.000	1.918	1.506	2.443
Outdoor activities (hours)							
1–2 vs. < 1	-0.327	0.061	28.541	<0.001	0.721	0.640	0.813
2–3 vs. < 1	-0.453	0.088	26.769	0.000	0.636	0.536	0.755
≥ 3 vs. < 1	-0.177	0.086	4.243	0.039	0.838	0.708	0.991
Sleep duration (hours)							
< 5 vs. ≥ 8	1.828	0.177	107.246	<0.001	6.224	4.403	8.797
5–6 vs. ≥ 8	1.519	0.097	247.297	<0.001	4.568	3.780	5.520
6–7 vs. ≥ 8	0.761	0.079	92.159	<0.001	2.141	1.833	2.501
7–8 vs. ≥ 8	0.323	0.083	15.319	<0.001	1.382	1.175	1.624

Notes *Nuclear family was considered as reference (ref). AOR: Adjusted odds ratio

**Fig. 2** Changes in the detection rate of depressive symptoms among participants**Fig. 3** Changes in the detection rate of possible and definite depressive symptoms among participants**Table 6** Logistic regression analysis of depressive symptoms detection rates (with 2019 as the reference year) ($N=9831$)

Year	Middle school		Junior high school		Total	
	β	AOR(95%CI)	β	AOR(95%CI)	β	AOR(95%CI)
2019		1.00(ref)		1.00(ref)		1.00(ref)
2020	0.297	1.346(1.092–1.660)	-0.311	0.732(0.584–0.919)	0.015	1.015(0.871–1.183)
2021	0.341	1.407(1.129–1.753)	-0.101	0.904(0.720–1.136)	0.111	1.117(0.954–1.309)
2022	-0.261	0.770(0.610–0.972)	-0.282	0.754(0.601–0.946)	-0.270	0.764(0.649–0.898)
2023	-0.203	0.816(0.651–1.024)	-0.223	0.800(0.642–0.997)	-0.211	0.810(0.692–0.947)

Notes AOR: Adjusted odds ratio

Table 7 Logistic regression analysis of depressive symptoms detection rates (with 2021 as the reference year) (N = 9831)

Years	Middle school		Junior high school		Total	
	β	AOR(95%CI)	β	AOR(95%CI)	β	AOR(95%CI)
2019	-0.341	0.711(0.570–0.886)	0.101	1.106(0.880–1.390)	-0.111	0.895(0.764–1.048)
2020	-0.044	0.957(0.773–1.184)	-0.211	0.810(0.638–1.028)	-0.096	0.908(0.775–1.064)
2021		1.00(ref)		1.00(ref)		1.00(ref)
2022	-0.602	0.548(0.432–0.694)	-0.182	0.834(0.657–1.058)	-0.381	0.683(0.578–0.808)
2023	-0.544	0.580(0.461–0.730)	-0.122	0.885(0.703–1.115)	-0.322	0.725(0.616–0.852)

Notes AOR: Adjusted odds ratio

in other studies [22, 23], which may be owing to differences in sampling scope and choice of the assessment scale used. A study on Chinese adolescents administered the 9-item Patient Health Questionnaire (PHQ-9) and revealed that approximately 15.8% reported depressive symptoms in 2021 [24]. The CES-D scale may have had a higher rate of false positives.

We found that sex, age, sleep duration, and duration of outdoor activities were factors that influenced middle and junior high school students' depressive symptoms, which was consistent with Qi et al.'s results [23], who used snowball sampling in China. Generally, girls are more sensitive than boys and more affected by the negative impacts of changes in the social environment; thus, girls may have higher levels of depression [15]. Most Chinese students experience high academic pressure [25], and junior high school students experience even greater pressure than middle school students. Middle school students' limited psychological development makes them more susceptible to external pressures and personal developmental issues, which can lead to extreme negative emotions [26]. Subgroup logistic analysis indicated that among female middle school students, the risk of depressive symptoms was 1.393 times higher than among junior high school girls; the opposite was true for male middle school students. Among middle school students, being male was a protective factor, whereas in junior high school, this effect was not significant. Generally, girls in junior high school are aged between 12 and 15 years. Hormonal fluctuations in females during puberty and premenstrual periods may contribute to the onset of depressive symptoms [27]. Junior high school students tend to place greater emphasis on academic performance, which might lead them to consciously mask their depressive symptoms. This could result in a higher reported prevalence of depressive symptoms among middle school students compared with junior high school students. Sleep problems become particularly prominent during the later stages of adolescence. This may be associated with the fact that insufficient sleep during this period can result in increased psychological stress, decreased cognitive abilities, and greater emotional fluctuations, which can severely affect mental health [28]. During the pandemic, school closures, home-based online learning,

reduced sleep restrictions, prolonged screen exposure, and decreased duration of outdoor activities may have led to sleep disorders [29].

To the best of our knowledge, this study is the first to explore the changes and trends in depressive symptoms among Chinese adolescents before and after the COVID-19 pandemic. Our study revealed that the proportion of depressive symptoms among middle school students increased in 2020 and 2021 and subsequently significantly declined in 2022 and 2023, which presented an inverted U-shaped trend. This trend was not observed among junior high school students, as the overall detection rate of depressive symptoms exhibited a declining trend. Several studies have indicated an increase in depression among adolescents during the early stages of the COVID-19 pandemic [15, 21, 30, 31]; however, research on changes in adolescent depression during its later stages is limited. Yang et al.'s ecological study [14] in Shandong, which included 1,170 students, revealed a downward trend in depression among adolescents during the later stages of the pandemic; they focused on middle school students, and their result was consistent with ours. In the 2022 Adolescent Mental Health Status Survey Report, the proportion of adolescents at risk of depression (14.8%) was lower compared with 2020 (19.0%) [6], which was also consistent with our findings.

Compared with 2019, the detection rate of depressive symptoms among middle and junior high school students was low in 2023. This suggests that the COVID-19 pandemic did not have long-term adverse effects on their mental health and may have even reduced the detection rates of depressive symptoms; however, this positive effect is not significant among students who definitely exhibit depressive symptoms. Individuals with strong social support and timely psychological interventions can experience positive psychological growth to an extent after experiencing stressful events [32]; this may explain the decreased proportion of possible depressive symptoms among middle school students. Additionally, adolescents who recovered from depression after the pandemic may be more sensitive to the needs of others and may have an increased prosocial tendency [14]. A study conducted in Shandong found that the prevalence of adolescent obesity exhibited an increasing trend

followed by a decreasing trend from 2019 to 2022. This change may also be attributed to a lack of outdoor activities. They believe that the adaptability of adolescents has been enhanced after the pandemic [33]. Decline in depressive symptoms levels among junior high school students after 2021 may also be related to the issuance of the “Opinions on Further Reducing the Homework Burden and After-school Training Burden for Students in Compulsory” by the General Office of the Central Committee of the Communist Party of China and State Council [34]. Reduced homework and off-campus training burden may increase students’ sleep duration and time spent on outdoor activities. A Korean longitudinal survey of 1185 girls and 1403 boys found a causal relationship between daily smartphone use time and risk of developing depressive symptoms and suicidal thoughts; however, this relationship was only significant in girls [35]. This suggested another possible explanation: during the COVID-19 pandemic, China implemented online classes, which inevitably led to an increase in teenagers’ daily exposure to smartphones. However, McColl et al. stated that the consequences of the COVID-19 pandemic will continue to impact children’s health and development in the coming years [36]; this may explain why the detection rate of students with depressive symptoms has not significantly changed.

A cross-sectional study conducted in Melbourne, Australia, investigated 741 middle school students and found that sleep quality significantly mediated the relationship between age and depressive symptoms [28]. In our research findings, middle school and junior high school students exhibited significant differences in the prevalence of depressive symptoms. Furthermore, they also exhibited variations in the trends of depressive symptom development. Differences between junior high and middle school students are reflected in age and academic pressure. Adolescents in middle school are younger and experience relatively less academic pressure; thus, their mental health is more dependent on family atmosphere and interactions with peers. Studies have established that academic pressure is negatively correlated with students’ physical and mental health; excessive academic pressure is considered a significant risk factor that can impair adolescents’ physical and mental well-being [26], which can make students more susceptible to negative emotions, such as anxiety, depression, and irritability [37]. In our study, no significant trend was observed in the detection rate of depressive symptoms among junior high school students from 2019 to 2023. During the COVID-19 pandemic, the detection rate of depressive symptoms among junior high school students even appeared to decline. This may be attributed to the fact that academic pressure is a primary reason for the onset of depressive moods among junior high school students. Furthermore,

the prevalence of the pandemic did not alter this academic pressure, which is largely driven by the college entrance examination. Meanwhile, school lockdowns and class suspensions as prevention measures might have increased the rest and sleep time for junior high school students, thereby alleviating depressive symptoms.

Our study possesses certain advantages and limitations. The large sample size enhanced the robustness of our research findings. Stratified random cluster sampling enabled us to include all students from the selected classes, which ensured a comprehensive representation. The cooperation of schools, participants’ high compliance and examination rates, and trained investigators further strengthened the reliability and validity of our results. Multivariate analysis and stratified analysis further controlled for confounding factors. However, our study has certain limitations. First, this was a cross-sectional study. It did not involve the long-term follow-up of a fixed population, which limits our ability to infer causal relationships. Therefore, our results require cautious interpretation. Future studies should utilize longitudinal data with extended follow-up periods to explore the long-term trends in mental health issues among adolescents. Second, our findings were largely based on self-reported data from participants. Although the investigators emphasized confidentiality and that results would only be used for scientific research at the beginning of the survey, this may still lead to certain biases. Third, the CES-D was used solely as a screening tool and cannot be used for diagnostic purposes, which may overestimate the prevalence of depressive symptoms. Additionally, our participants were mainly from Nantong in Jiangsu Province. Therefore, the sample does not fully represent the general Chinese adolescent population. Significant differences in education and environment across China require further research to include more provinces and participants. Our study highlights the necessity of addressing the relationship between lifestyle and depressive symptoms in the development of cognitive and behavioral programs aimed at improving adolescents’ mental health. Focusing on adolescents during periods of significant environmental changes is crucial, as the impact of these changes on their emotional and psychological well-being is often overlooked. Timely and effective mitigation of these impacts represents a key area for future research and intervention efforts.

Conclusions

Factors influencing depressive symptoms among middle and junior high school students include educational stage, sex, family type, duration of outdoor activities, and duration of sleep. We found that the proportion of possible depressive symptoms among middle school students exhibited a trend of first rising and subsequently falling

from 2019 to 2023; junior high school students were less affected by the COVID-19 and exhibited no significant trend. Compared with 2019, the proportion of depressive symptoms was lower in 2023, after controlling for influencing factors. The COVID-19 pandemic did not seem to have long-term adverse effects on middle and junior high school students' depressive symptoms. However, we still need to focus on students exposed to risk factors, such as being female or coming from a restructured family. We recommend that early intervention should be promptly initiated to mitigate the impact of environmental stress and prevent the further development of depressive tendencies among students, especially in the context of responding to the impacts of sudden social changes. Further longitudinal studies are required to obtain more information.

Abbreviations

CES-D Center for Epidemiologic Studies Depression Scale

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12889-024-21252-8>.

Supplementary Material 1: Supplementary Table 1. Sex and educational stage composition ratios from 2019 to 2023.

Supplementary Material 2: Supplementary Table 2. Multivariate logistic analysis of depressive symptoms by sex.

Supplementary Material 3: Supplementary Table 3. Multivariate logistic analysis of depressive symptoms by educational stage.

Supplementary Material 4: Additional file 4. Strobe guidelines for the manuscript.

Acknowledgements

The authors extend their gratitude to all members of the Nantong Center for Disease Control and Prevention for their commitment to this work.

Author contributions

XZ: Writing – original draft. NA: Data curation, Writing – review & editing. ZT: Design, writing – review & editing. JH: Data curation, Writing – review & editing. QR: Data curation. YW: Data curation. All authors reviewed the manuscript.

Funding

This study was supported by the National Natural Science Foundation of China [82373688, 81773541], Jiangsu Province Engineering Research Center of Molecular Target Therapy and Companion Diagnostics in Oncology (SGK1202407), the Priority Academic Program Development of Jiangsu Higher Education Institutions at Soochow University, and the Nantong Health Commission (NO. MS2022088). The funding bodies had no role in the design or execution of this research.

Data availability

The data of this study are available from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate

This study was approved by the Ethics Committee of the Nantong City Disease Prevention and Control Center (approval no. 2023004). All

methods were performed according to the Declaration of Helsinki and its future amendments. Written informed consent was obtained from all the participants and their parents or caregivers.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

Author details

¹Nantong Center for Disease Control and Prevention, Nantong, China

²Department of Biostatistics, School of Public Health, Jiangsu Key Laboratory of Preventive and Translational Medicine for Geriatric Diseases, MOE Key Laboratory of Geriatric Diseases and Immunology, Suzhou Medical College of Soochow University, Suzhou 215123, Jiangsu, P.R. China

Received: 3 July 2024 / Accepted: 30 December 2024

Published online: 02 January 2025

References

1. UNICEF. Adolescent mental health. 2020. <https://www.unicef.cn/media/20001/file/ADOLESCENT%20MENTAL%20HEALTH.pdf>. Accessed 7 Feb 2024.
2. GBD 2019 Diseases and Injuries Collaborators. Global burden of 369 diseases and injuries in 204 countries and territories, 1990–2019: a systematic analysis for the global burden of Disease Study 2019. *Lancet*. 2020;396(10258):1204–22.
3. Bitsko RH, Holbrook JR, Ghandour RM, Blumberg SJ, Visser SN, Perou R, et al. Epidemiology and impact of health care provider-diagnosed anxiety and depression among us children. *J Dev Behav Pediatr*. 2018;39(5):395–403.
4. Harris TB, Udoetuk SC, Webb S, Tatem A, Nutile LM, Al-Mateen CS. Achieving mental health equity: children and adolescents. *Psychiatr Clin North Am*. 2020;43(3):471–85.
5. Pozuelo JR, Desborough L, Stein A, Cipriani A. Systematic review and meta-analysis: depressive symptoms and risky behaviors among adolescents in low- and middle-income countries. *J Am Acad Child Adolesc Psychiatry*. 2022;61(2):255–76.
6. Psychological China, National Mental Health Development Report. 2022. 2023. http://psy.china.com.cn/node_1013714.htm. Accessed 10 March 2024.
7. Akande OW, Akande TM. COVID-19 pandemic: a global health burden. *Niger Postgrad Med J*. 2020;27(3):147–55.
8. Bourmistrova NW, Solomon T, Braude P, Strawbridge R, Carter B. Long-term effects of COVID-19 on mental health: a systematic review. *J Affect Disord*. 2022;299:118–25.
9. Ahmed GK, Khedr EM, Hamad DA, Meshref TS, Hashem MM, Aly MM. Long term impact of COVID-19 infection on sleep and mental health: a cross-sectional study. *Psychiatry Res*. 2021;305:114243.
10. Singh S, Roy D, Sinha K, Parveen S, Sharma G, Joshi G. Impact of COVID-19 and lockdown on mental health of children and adolescents: a narrative review with recommendations. *Psychiatry Res*. 2020;293:113429.
11. Xie X, Xue Q, Zhou Y, Zhu K, Liu Q, Zhang J, et al. Mental health status among children in home confinement during the coronavirus disease 2019 outbreak in Hubei Province, China. *JAMA Pediatr*. 2020;174(9):898–900.
12. Xiong J, Lipsitz O, Nasri F, Lui L, Gill H, Phan L, et al. Impact of COVID-19 pandemic on mental health in the general population: a systematic review. *J Affect Disord*. 2020;277:55–64.
13. Golberstein E, Wen H, Miller BF. Coronavirus disease 2019 (COVID-19) and mental health for children and adolescents. *JAMA Pediatr*. 2020;174(9):819–20.
14. Yang Z, Peng H, Xin S. A longitudinal study on depression and anxiety among Chinese adolescents in the late phase of the COVID-19 pandemic: the trajectories, antecedents, and outcomes. *Acta Physiol Sinica*. 2024;56.
15. Ma L, Mazidi M, Li K, Li Y, Chen S, Kirwan R, et al. Prevalence of mental health problems among children and adolescents during the COVID-19 pandemic: a systematic review and meta-analysis. *J Affect Disord*. 2021;293:78–89.
16. Radloff LS. The use of the Center for Epidemiologic Studies Depression Scale in adolescents and young adults. *J Youth Adolesc*. 1991;20(2):149–66.

17. Verhoeven M, Sawyer MG, Spence SH. The factorial invariance of the CES-D during adolescence: are symptom profiles for depression stable across gender and time? *J Adolesc*. 2013;36(1):181–90.
18. Chen ZY, Yang XD, Li XY. Psychometric features of CES-D in Chinese adolescents. *Chin J Clin Psychol*. 2009;17:443–8.
19. Ling Y, Wei Y, Yi JY, Xiao J, Yao SQ. Factorial structure of the CES-D scale among Chinese high school students. *Chin J Clin Psychol*. 2008;16:265–7.
20. Wang M, Armour C, Wu Y, Ren F, Zhu X, Yao S. Factor structure of the CES-D and measurement invariance across gender in Mainland Chinese adolescents. *J Clin Psychol*. 2013;69(9):966–79.
21. Duan L, Shao X, Wang Y, Huang Y, Miao J, Yang X, et al. An investigation of mental health status of children and adolescents in China during the outbreak of COVID-19. *J Affect Disord*. 2020;275:112–8.
22. Bin X, Qu KY, Wang YH, Chen L, Xiong YJ, Wen JF, et al. Prevalence of depression, anxiety in China during the COVID-19 pandemic: an updated systematic review and meta-analysis. *Front Public Health*. 2023;11:1267764.
23. Qi H, Liu R, Feng Y, Luo J, Lei SM, Cheung T, et al. Prevalence of depression and its associated factors among adolescents in China during the early stage of the COVID-19 outbreak. *PeerJ*. 2021;9:e12223.
24. Cheng H, Hu W, Luo S, Feng X, Chen Z, Yu X et al. Pathways linking loneliness and depressive symptoms among Chinese adolescents: the mediating role of sleep disturbance. *J Affect Disord*. 2025;370:235–40.
25. Högberg B. Educational stressors and secular trends in school stress and mental health problems in adolescents. *Soc Sci Med*. 2021;270:113616.
26. Bazarova NN, Choi YH, Whitlock J, Cosley D, Sosik V. Psychological distress and emotional expression on Facebook. *Cyberpsychol Behav Soc Netw*. 2017;20(3):157–63.
27. Albert PR. Why is depression more prevalent in women? *J Psychiatry Neurosci*. 2015;40(4):219–21.
28. Raniti MB, Allen NB, Schwartz O, Waloszek JM, Byrne ML, Woods MJ, et al. Sleep duration and sleep quality: associations with depressive symptoms across adolescence. *Behav Sleep Med*. 2017;15(3):198–215.
29. Liu Z, Tang H, Jin Q, Wang G, Yang Z, Chen H, et al. Sleep of preschoolers during the coronavirus disease 2019 (COVID-19) outbreak. *J Sleep Res*. 2021;30(1):e13142.
30. Zhou SJ, Zhang LG, Wang LL, Guo ZC, Wang JQ, Chen JC, et al. Prevalence and socio-demographic correlates of psychological health problems in Chinese adolescents during the outbreak of COVID-19. *Eur Child Adolesc Psychiatry*. 2020;29(6):749–58.
31. Hawes MT, Szenczy AK, Klein DN, Hajcak G, Nelson BD. Increases in depression and anxiety symptoms in adolescents and young adults during the COVID-19 pandemic. *Psychol Med*. 2022;52(14):3222–30.
32. Zhang N, Yang S, Jia P. Cultivating Resilience during the COVID-19 pandemic: a socioecological perspective. *Annu Rev Psychol*. 2022;73:575–98.
33. Qin G, Li H, Qin Y, Liu B. Is the COVID-19 epidemic affecting the body mass of Chinese teenagers? - a longitudinal follow-up study. *Front Public Health*. 2023;11:1269076.
34. The General Office of the CPC Central Committee and the General Office of the State Council. Opinions on further reducing the homework burden and after-school training burden for students in compulsory. 2021. http://www.moe.gov.cn/jyb_xgk/moe_1777/moe_1778/202107/t20210724_546576.html. Accessed 7 Feb 2024.
35. Rudolf R, Kim N. Smartphone use, gender, and adolescent mental health: longitudinal evidence from South Korea. *SSM Popul Health*. 2024;28:101722.
36. McColl R, Nescott E. Youth Mental Health in the wake of the COVID-19 pandemic. *Dela J Public Health*. 2024;10(4):62–6.
37. Zhang WJ, Yan C, Shum D, Deng CP. Responses to academic stress mediate the association between sleep difficulties and depressive/anxiety symptoms in Chinese adolescents. *J Affect Disord*. 2020;263:89–98.

Publisher's note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.